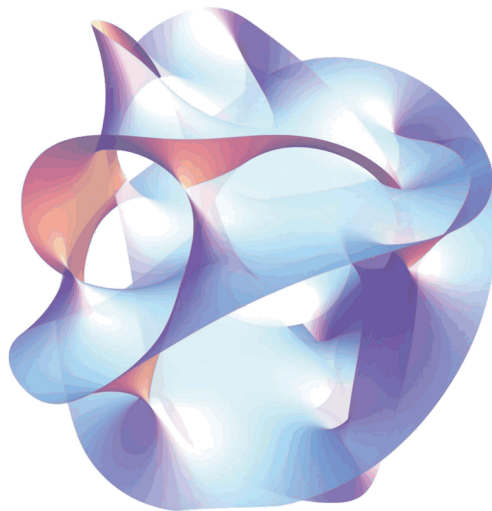


Dimensional reduction in supergravity and duality symmetries

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Notations and abbreviations

We will during this internship use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices ($\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots$)
- Latin hatted letter for flat indices ($\hat{p}, \hat{m}, \hat{n}, \dots$)
- For space-time indices taking D values, late greek letters are used in the curved case ($\mu, \nu, \rho, \dots$) and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case ($\alpha, \beta, \gamma, \dots$) and Latin letters in the flat case (a,b,c)

The general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate x^{μ} and internal coordinate y^{α} . The flat lorenz

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1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Indeed its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything other force. A theory to solve this issue is to add a new symmetry to our universe, the super-symmetry. This new symmetry is mainly used in two theory, the first one is supergravity which as the name suggest is a theory which try to use the principle of super-symmetry with general relativity. The other big theory is the Superstring theory which try to do the same thing as the last one but with string theory. An advantage of the first one over the later is that supergravity does not demand supplementary dimension, which Superstring demands. Here we will do an amalagation of both, we will add more dimensino while not going to close to the string theory. In this internship we will begin by describing what is a field theory to seek how to describe the different action of the field we will be using, after that we will to follow the step of [1] by doing a dimensional reduction from D+E dimension to 4 dimension. However following the step [3] of we will not juste use the classical Einstein action, but the NS-NS action of super gravity which basically add two new field. We will do this by using the vielbein formalism as it is useful for the description of fermion even here we will solely try to describe bosonics fields. Then we will see that even if in high dimension only the $SL(E,E)$ dimension is detectable without to much work, that when we go into low dimension another $O(E,E)$ symmetry will appear, and that mean that a lot of other theory are in fact equivalent to the one we are using. Finally will suppose that none of the field depend on the internal coordinate, therefor any field we will have $\partial_\alpha A^\mu = 0$

2 spin connexion, vielbein and flatspace

In general relativity we often encounter difficult to handle metrics such as the Kerr-Newman metric. A good idea we can have is to then work in an orthonormal basis $\{e_\mu^a\}$, which follow:

$$g_{\mu\nu} V_a^\mu V_b^\nu = \eta_{ab} \quad (1)$$

where g is the original metric and η is the Minkowski metric.

We call this basis the vielbein. Of course we can define the dual of the vielbein otherwise called the inverse vielbein with

$$V_\nu^a V_b^\nu = \delta_b^a \quad (2)$$

and of course that mean that we can go back with to the original metric with

$$g_{\mu\nu} = \eta_{ab} V_\mu^a V_\nu^b \quad (3)$$

Of course we can also change the basis any random vector, form or tensor with this, the method is to simply do

$$T^a = V_\mu^a T^\mu \quad (4)$$

$$T_a = V_a^\mu T_\mu \quad (5)$$

$$T_b^a = V_\mu^a V_b^\nu T_\nu^\mu \quad (6)$$

Now of course other cange need to happen to the classical transformation for this to work, such as the classical christofells connexion which does not work anymore in this context so we introduce the spin(or latin) connexion whith

$$\nabla_\mu X_b^a = \partial_\mu X_b^a + \omega_\mu^a{}_c X_b^c - \omega_\mu^c{}_b X_c^a \quad (7)$$

Of course we can still make link with the levi-civita connexion with the equations

$$\Gamma_{\mu\lambda}^\nu = V_a^\nu \partial_\mu V_\lambda^a + V_a^\nu V_\lambda^b \omega_\mu^a{}_b, \quad (8)$$

$$\omega_\mu^a{}_b = V_\nu^a V_b^\lambda \Gamma_{\mu\lambda}^\nu - V_b^\lambda \partial_\mu V_\lambda^a, \quad (9)$$

With those relation we can do prove the anti-symetrie of the last two index of the spin connexion

$$\begin{aligned} \nabla_\mu \eta_{ab} &= \partial_\mu X_{ab} - \omega_\mu^a{}_c \eta_{cb} - \omega_\mu^c{}_b \eta_{ac} \\ &= -\omega_{\mu ab} - \omega_{\mu ba} \\ &= 0 \end{aligned} \quad (10)$$

so in effect we have

$$-\omega_{\mu ab} = \omega_{\mu ba} \quad (11)$$

With all of that said we can now write down the rule to lower and raise index. For exemple in

3 Lagrangian field theory

3.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantities called Lagrangian density, which will often referred to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dx dy dz \quad (12)$$

Now using esintein convention we can rewritte the action in a new light :

$$S = \int \mathcal{L} dx^\mu \quad (13)$$

3.2 General exemple for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical description any field (often the simpliest). For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.

So for a scalar field we have :

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi \quad (14)$$

and for a vectorial field we have

$$\mathcal{L} = -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} \quad (15)$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A^\nu - \partial_\nu A^\mu$$

In a vaccum the einstein equation give us

$$\mathcal{L} = R$$

where R is ricci scalar finally for the lagrangian of a two form fields whe will look in [3] where it is given by the equation :

$$\mathcal{L} = \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'} \quad (16)$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + cyc.perm$$

3.3 NS-NS Action Of supergravity

With those object we can effectively create the NS-NS action which has this form:

$$S = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{\hat{\phi}} (\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} - \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'}) \quad (17)$$

where $\rho(E)$ is a normalisation factor, corresponding in practice to the invariant volume of the internal space.

And where $\hat{R}$ is similar as the classical Hilbert-Einstein action but in $D+E$ dimension, $\hat{\phi}$ is a dilaton field and $-\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho}$ is the action of a two form. Finally as no field depend on internal component and knowing what $\rho(E)$ we can just simplify this integral. Now we will separate this action into 3 part one for each action so we have Now that we have this action in $D+E$ dimension we need to in the same way as in the [1] reduce it, which basically mean look on how would a D dimensional observer would perceives those extra dimension.

3.4 different part of the action

We can separate the action into two main component one being the gravitational part with the dilaton and the otherone being the two forms part so we have:

$$S_{\hat{g}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} [\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi}], \quad (18a)$$

$$S_{\hat{B}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho}, \quad (18b)$$

4 gravitational action reduction

We will begin with the gravitational part of the NS-NS action.

4.1 Rewriting the action

For this part we don't need the $\hat{g}^{\hat{\mu}\hat{\nu}}\partial_{\hat{\mu}}\hat{\phi}\partial^{\hat{\nu}}\hat{\phi}$ part of the action so we will fully ignore it. Now using the fact that $\hat{R}$ can be written as

$$\hat{R} = \hat{V}^{\hat{\mu}\hat{r}}\hat{V}^{\hat{\nu}\hat{s}}\hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}}$$

where

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} = \partial_{\hat{\mu}}\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - (\mu \leftrightarrow \nu)$$

and the ω are defined By

$$\hat{\omega}_{trs} = -\hat{\Omega}_{t\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}t} - \hat{\Omega}_{\hat{s}t\hat{r}}$$

and finally the Ω are defined thanks to the vielbein with

$$\hat{\Omega}_{\hat{r}\hat{s}t} = \frac{1}{2}(\hat{V}_{\hat{r}}^{\hat{\mu}}\hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{r}}^{\hat{\nu}}\hat{V}_{\hat{s}}^{\hat{\mu}})\partial_{\hat{\nu}}\hat{V}_{\hat{\mu}\hat{r}}$$

with all of that we are going to rewrite the action $S_{\hat{g}}$ in terms of $\hat{\omega}$

$$\begin{aligned} S_{\hat{g}} &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{R} \\ &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} \\ &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} ([\partial_{\hat{\mu}}\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} - \partial_{\hat{\nu}}\hat{\omega}_{\hat{\mu}\hat{r}\hat{s}}] + [\hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{\mu}\hat{t}\hat{s}}]) \\ &= \int dx^D \{2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}}\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} - \hat{\omega}_{\hat{r}}^{\hat{s}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}}]\} \\ &= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} - \hat{\omega}_{\hat{r}}^{\hat{s}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}}\partial_{\hat{\mu}}(\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}})\} \\ &= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} - \hat{\omega}_{\hat{r}}^{\hat{s}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}}[\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \partial_{\hat{\mu}}\hat{V}^{\hat{\nu}\hat{s}} + \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}}\hat{V}^{\hat{\mu}\hat{r}} + e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}}\hat{V} + \hat{V} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}}e^{-\hat{\phi}}]\} \\ &= \int dx^D \hat{V} e^{-\hat{\phi}} \{\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{r}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} + \hat{\omega}_{\hat{r}}^{\hat{s}}\hat{\omega}_{\hat{s}}^{\hat{t}}\hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} + 2\hat{\omega}_{\hat{s}}^{\hat{r}}\hat{V}_{\hat{r}}^{\hat{\mu}}\partial_{\hat{\mu}}\hat{\phi}\} \end{aligned} \quad (19)$$

4.2 Vielbein Anstaz

Now that we have explicitly written the action in terms of either latin connexion and or the two other fields, what we have to do is to suppose an ansatz for the vielbein. Why the vielbein and not the metric you may ask? Because as stated in [A](#), all the information of the metric can be contained within the vielbein, and as we primarily work with them for the action, it is easier to do an ansatz on them. Now we will do the same ansatz as in the [\[1\]](#), however we will consider that γ they use is null since the beginning, and we will use the same parameter number as in [\[3\]](#). Finally thanks to the local lorentz invariance we can choose a triangular parametrisation giving us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\hat{\mu}}^r & A_{\hat{\mu}}^{(1)\beta}\Phi_{\beta}^a \\ 0 & \Phi_{\alpha}^a \end{pmatrix} \quad (20)$$

Now still using [\(60\)](#) we can compute the inverse vielbein and then the metric and then the inverse metric that gave us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_r^{\mu} & -A_r^{\alpha} \\ 0 & \Phi_a^{\alpha} \end{pmatrix}, \quad (21)$$

for the inverse vielbein, and

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} - A_{\mu}^{\lambda}A_{\nu\lambda} & -A_{\mu\beta} \\ A_{\nu\alpha} & -h_{\alpha\beta} \end{pmatrix}, \text{ and } \hat{g}^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g^{\mu\nu} & -A^{\mu\beta} \\ -A^{\nu\alpha} & -h^{\alpha\beta} + A_{\rho}^{\beta}A^{\rho\alpha} \end{pmatrix}, \quad (22)$$

for the metric

4.3 Gauge observation

Now that we have done the anstaz, we would like to know what each object are, for that we can use the classical gauge transformation $x'^\mu = x^\mu + \xi^\mu(x)$, and use the fact that it can be easily rewritten as a Lie derivative. Indeed we already know of different object transform under Lie derivative so we would easily be able indentify what each object are. So firstly let's list every known transformation:

$$\delta\phi(x) = \xi^\mu \partial_\mu \Phi(x), \quad (23)$$

$$\delta U^\mu = \xi^\rho \partial_\rho U^\mu - (\partial_\rho \xi^\mu) U^\rho, \quad (24)$$

$$\delta\omega_\mu = \xi^\rho \partial_\rho \omega_\mu + (\partial_\rho \xi^\mu) \omega_\rho, \quad (25)$$

$$\delta T_\nu^\mu = \xi^\rho \partial_\rho T_\nu^\mu - (\partial_\rho \xi^\mu) T_\nu^\rho + (\partial_\nu \xi^\rho) T_\rho^\mu, \quad (26)$$

we now apply the form transformation to each block of the vielbein, in order to see whether it corresponds to some familiar type of field still thanks to (23). Let us start with the block $\hat{V}_\mu^r$:

$$\begin{aligned} \delta \hat{V}_\mu^r &= \delta V_\mu^r = \xi^\rho \partial_\rho \hat{V}_\mu^r + \partial_\mu \xi^\rho \hat{V}_\rho^r \\ &= \xi^\rho \partial_\rho \hat{V}_\mu^r + \partial_\mu \xi^\nu \hat{V}_\nu^r \\ &= \xi^\rho \partial_\rho V_\mu^r + \partial_\mu \xi^\nu V_\nu^r \end{aligned}$$

which is exactly the transformation law of a vector field under a Lie derivative [4]. Applying the same method to the other blocks, we obtain, for Φ_α^a :

$$\delta \Phi_\alpha^a = \xi^\rho \partial_\rho \Phi_\alpha^a$$

which is exactly the transformation law of a scalar field. Finally, for A_μ^α we obtain:

$$A_\mu^\alpha = \xi^\rho \partial_\rho A_\mu^\alpha + \partial_\mu \xi^\rho A_\rho^\alpha + \partial_\mu \xi^\alpha$$

which corresponds to the transformation law of a vector field, up to an extra correction term. One may further note that, restricting to ξ^α only, we obtain

$$\delta V_\mu^r = \delta \Phi_\alpha^a = 0, \quad (27a)$$

$$\delta A_\mu^\alpha = \partial_\mu \xi^\alpha. \quad (27b)$$

so that what remains of the D -dimensional diffeomorphisms, once restricted to ξ^α independent of the internal coordinates y , corresponds to a set of E abelian gauge symmetries. The vielbein in (20) thus contains the D -dimensional vielbein (i.e. the graviton), E vector fields A_μ^α , and E^2 scalar fields Φ_α^a . However, because of the local Lorentz invariance in the internal directions, only $\frac{1}{2}E(E+1)$ of these scalars actually correspond to propagating degrees of freedom. We also introduce $\delta = \det \Phi$

4.4 Ricci scalar reduction

Now that we better know the tool we will use, We can begin the calculation. For a starter we will do the Einstein action in D+E dimension twiked by the addition of the dilation.

Firstly let's begin by expending completly the $\hat{S}_g$, this will show us that we technically have to compute 8 ω , however thanks to the antisymmetry og the last to index we can reduce that to 6. So the action is

$$\begin{aligned} S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \Big[& \hat{\omega}_{bc}^a \hat{\omega}_{a}^{bc} - \hat{\omega}_{tb}^a \hat{\omega}_a^{bt} + \hat{\omega}_{sc}^a \hat{\omega}_a^{sc} + \hat{\omega}_{st}^a \hat{\omega}_a^{st} - \hat{\omega}_{bc}^r \hat{\omega}_r^{bc} - \hat{\omega}_{tb}^r \hat{\omega}_r^{bt} - \hat{\omega}_{sc}^r \hat{\omega}_r^{sc} \\ & \hat{\omega}_{st}^r \hat{\omega}_r^{st} + \hat{\omega}_{ac}^a \hat{\omega}_b^{bc} + \hat{\omega}_{ta}^a \hat{\omega}_b^{bt} + \hat{\omega}_{ac}^a \hat{\omega}_s^{sc} - \hat{\omega}_{ta}^a \hat{\omega}_s^{st} + \hat{\omega}_{rc}^r \hat{\omega}_b^{bc} - \hat{\omega}_{rt}^r \hat{\omega}_b^{bt} + \hat{\omega}_{rc}^r \hat{\omega}_s^{sc} + \hat{\omega}_{rt}^r \hat{\omega}_s^{st} + \\ & 2\hat{\omega}_s^{rs} \hat{V}_r^\mu \partial_\mu \hat{\phi} + 2\hat{\omega}_b^{rb} \hat{V}_r^\mu \partial_\mu \hat{\phi} \Big] \end{aligned} \quad (28)$$

It then becomes clear that we only need to compute six components of ω , namely

$$\begin{aligned} & \hat{\omega}_{pmn}, \hat{\omega}_{pma}, \hat{\omega}_{pab}, \\ & \hat{\omega}_{cmn}, \hat{\omega}_{cma}, \hat{\omega}_{cab}, \end{aligned}$$

One can also check that computing these six components only requires the following eight components of Ω :

$$\begin{aligned} & \hat{\Omega}_{pmn}, \\ & \hat{\Omega}_{pma}, \hat{\Omega}_{pam}, \hat{\Omega}_{apm}, \\ & \hat{\Omega}_{mab}, \hat{\Omega}_{amb}, \hat{\Omega}_{abm}, \\ & \hat{\Omega}_{abc}, \end{aligned}$$

Using (21), (4.1) and (4.1), we can now set out a clear strategy: first the Ω 's, and then the ω 's. Now let's begin the computation of the Ω 's. Starting with $\hat{\Omega}_{pmn}$:

$$\begin{aligned}
\hat{\Omega}_{pmn} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_m^{\nu} - \hat{V}_m^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}n} \\
&= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_m^{\nu} - \hat{V}_m^{\mu}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\mu n} \\
&= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_m^{\nu} - \hat{V}_m^{\mu}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\mu n} \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n})
\end{aligned}$$

For $\hat{\Omega}_{pma}$, we find:

$$\begin{aligned}
\hat{\Omega}_{pma} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_m^{\nu} - \hat{V}_m^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}a} \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(A_{\mu}^{\alpha}\Phi_{a\alpha}) + (A_p^{\alpha}V_m^{\nu} - \hat{A}_m^{\alpha}V_p^{\nu})\partial_{\nu}(\Phi_{a\alpha}) \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_{\nu}(A_{\mu}^{\alpha})\Phi_{a\alpha} + \partial_{\nu}(\Phi_{a\alpha})A_{\mu}^{\alpha}) + \kappa 1(A_p^{\alpha}V_m^{\nu} - \hat{A}_m^{\alpha}V_p^{\nu})\partial_{\nu}(\Phi_{a\alpha}) \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_{\nu}(A_{\mu}^{\alpha})\Phi_{a\alpha} + \partial_{\nu}(\Phi_{a\alpha})A_{\mu}^{\alpha}) + (A_p^{\alpha}V_m^{\nu} - \hat{A}_m^{\alpha}V_p^{\nu})\partial_{\nu}(\Phi_{a\alpha}) \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(A_{\mu}^{\alpha}\Phi_{a\alpha}) \\
&= (F^{\alpha}_{pm}\Phi_{a\alpha} - \partial_{\nu}(V_p^{\mu}V_m^{\nu})A_{\mu}^{\alpha}\Phi_{a\alpha} + \partial_{\nu}(V_p^{\nu}V_m^{\mu})A_{\mu}^{\alpha}\Phi_{a\alpha}) \\
&= F^{\alpha}_{pm}\Phi_{a\alpha}
\end{aligned}$$

For $\hat{\Omega}_{pam}$:

$$\begin{aligned}
\hat{\Omega}_{pam} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}m} \\
&= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_a^{\nu} - \hat{V}_a^{\mu}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\mu m} \\
&= 0
\end{aligned}$$

The same result holds for $\hat{\Omega}_{apm}$.

For $\hat{\Omega}_{pab}$:

$$\begin{aligned}
\hat{\Omega}_{pab} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{-1}{2}\hat{V}_a^{\hat{\mu}}V_p^{\nu}\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}\Phi_{\alpha b} \\
&= \frac{1}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}(h_{\alpha\beta}\Phi_b^{\beta}) \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{1}{2}\Phi_a^{\alpha}V_p^{\nu}h_{\alpha\beta}\partial_{\nu}\Phi_b^{\beta} \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{1}{2}\Phi_{\beta a}V_p^{\nu}\partial_{\nu}\Phi_b^{\beta} \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{1}{2}V_p^{\nu}\partial_{\nu}(\underbrace{\Phi_{\beta a}\Phi_b^{\beta}}_{=\delta_{ab} \text{ i.e is constant}}) - \Phi_b^{\beta}V_p^{\nu}\partial_{\nu}\Phi_{\beta a} \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} - \Phi_b^{\beta}V_p^{\nu}\partial_{\nu}\Phi_{\beta a}
\end{aligned}$$

For $\hat{\Omega}_{apb}$:

$$\begin{aligned}\hat{\Omega}_{apb} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu} - \hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}b} \\ &= \frac{1}{2}\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu}\partial_{\nu}\hat{V}_{\hat{\mu}b} \\ &= \frac{1}{2}\hat{V}_a^{\alpha}\hat{V}_p^{\nu}\partial_{\nu}\hat{V}_{\alpha b} \\ &= \frac{-1}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}\Phi_{\alpha b}\end{aligned}$$

For $\hat{\Omega}_{abp}$:

$$\begin{aligned}\hat{\Omega}_{abp} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_b^{\nu} - \hat{V}_b^{\hat{\mu}}\hat{V}_a^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}p} \\ &= 0\end{aligned}$$

The same holds for Ω_{abc} . Using (20), we can then obtain the ω 's:

$$\hat{\omega}_{pmn} = \omega_{pmn}, \quad (29a)$$

$$\hat{\omega}_{pma} = F_{pm}^{\alpha}\Phi_{\alpha a}, \quad (29b)$$

$$\hat{\omega}_{pab} = \frac{1}{2}\Phi_a^{\alpha}V_p^{\mu}\partial_{\mu}\Phi_{\alpha b} - (a \leftrightarrow b), \quad (29c)$$

$$\hat{\omega}_{cmn} = -F_{mn}^{\alpha}\Phi_{\alpha c}, \quad (29d)$$

$$\hat{\omega}_{cab} = 0 \quad (29e)$$

We can now conclude this part by just computing the multiplication and look at what we got. This give us.

$$\boxed{S = \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right.} \quad (30)$$

$$\left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} \right)$$

We can observe some interesting things. Firtstly we can see that we have refound the classical Einstein action, coupled with E electromagnetism field. So what we can conclude is that gravity in D+E dimation is gravity in 4d plus E electromagnetism field. and we find a few other field which will come in handy for the dilaton field reduction.

4.5 Adding the dilaton field reduction

For this part we will have to sum $S_{\hat{g}}$ and $S_{\hat{\phi}}$ that gave us

$$\begin{aligned}S_{\hat{g}+\hat{\phi}} &= \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right. \\ &\quad \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} \right)\end{aligned}$$

We then introduce the shifted dilaton as

$$\phi = \hat{\phi} - \frac{1}{2} \log h = \hat{\phi} - \log \delta \quad (31)$$

wich gave us

$$\boxed{S = \int dx^D V e^{-\phi} \left(R - \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right.} \quad (32)$$

$$\left. - g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \phi \partial_{\hat{\nu}} \phi \right)$$

5 Two form fiel reduction

5.1 The reduction

For the two form we will reduce it apart from the two other field, this those not cause any problem as the action is in effect linear.

We can rewrite (16) as

$$\mathcal{L}_{\hat{B}} = \frac{-1}{12} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \hat{H}^{\hat{\mu}\hat{\nu}\hat{\rho}}$$

here we use the trans

$$\begin{aligned}
&= \frac{-1}{12} \hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{V}_{\hat{\mu}}^{\hat{r}} \hat{V}_{\hat{\nu}}^{\hat{s}} \hat{V}_{\hat{\rho}}^{\hat{t}} \hat{H}^{\hat{\mu}\hat{\nu}\hat{\rho}} \\
&= \frac{-1}{12} (\hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}^{\hat{r}\hat{s}\hat{t}}) \\
&= \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + \hat{H}_{rsc} \hat{H}^{rsc} + \hat{H}_{rbt} \hat{H}^{rbt} + \hat{H}_{rbc} \hat{H}^{rbc} + \hat{H}_{ast} \hat{H}^{ast} + \hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abs} \hat{H}^{abs} + \hat{H}_{abc} \hat{H}^{abc})
\end{aligned}$$

however thanks to the fact that $\hat{B}_{\hat{\mu}\hat{\nu}}$ is antisymmetric one can easily show that this imply that $\hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is antisymmetric for any neighbor index. so we obtain

$$\mathcal{L}_{\hat{B}} = \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + 3\hat{H}_{rsc} \hat{H}^{rsc} + 3\hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abc} \hat{H}^{abc}) \quad (33)$$

and finally the action of B is written as

$$S_b = - \int dx^D \sqrt{-g} e^{-\phi} \left\{ \frac{1}{12} H_{\alpha\beta\gamma} H^{\alpha\beta\gamma st} + \frac{1}{4} H_{\mu\alpha\beta} H^{\mu\alpha\beta} + \frac{1}{4} H_{\mu\nu\alpha} H^{\mu\nu\alpha} + \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right\} \quad (34)$$

The absence of hat is due to the fact that we define low dimensional form as being:

$$T^\mu = V_r^\mu \hat{V}_\mu^r \hat{T}^{\hat{\mu}} \quad (35a)$$

$$T^\alpha = V_a^\alpha \hat{V}_\mu^a \hat{T}^{\hat{\mu}} \quad (35b)$$

$$T_\mu = V_\mu^r \hat{V}_r^\mu \hat{T}_{\hat{\mu}} \quad (35c)$$

$$T_\alpha = V_\alpha^a \hat{V}_a^\mu \hat{T}_{\hat{\mu}} \quad (35d)$$

$$(35e)$$

so with that we obtain (34)

Now we will compute each component of non hatted H in function of hatted H

to begin slowly as we have seen above $\hat{B}_{\alpha\beta} = B_{\alpha\beta}$ and if we use the y independence we get $H_{\alpha\beta\gamma} = 0$

We continue with $H_{\mu\alpha\beta}$

$$\begin{aligned}
H_{\mu\alpha\beta} &= V_\mu^r \hat{V}_r^\mu \hat{H}_{\hat{\mu}\alpha\beta} \\
&= \partial_\mu B_{\alpha\beta}
\end{aligned} \quad (36)$$

those two were quite easy however the next two are going to be more troublesome, but we are going to introduce new object Now for $H_{\mu\nu\beta}$

$$\begin{aligned}
H_{\mu\alpha\beta} &= V_\mu^r V_\nu^s \hat{V}_r^\mu \hat{V}_s^\nu \hat{H}_{\hat{\mu}\hat{\nu}\alpha} \\
&= \hat{H}_{\mu\nu\alpha} - (A_\mu^{(1)\alpha} \hat{H}_{\alpha\nu\rho} + A_\rho^{(1)\alpha} \hat{H}_{\mu\alpha\nu}) \\
&= \partial_\mu \hat{B}_{\nu\alpha} + \partial_\nu \hat{B}_{\alpha\mu} - A_\mu^{(1)\beta} \partial_\nu \hat{B}_{\alpha\beta} - A_\nu^{(1)\beta} \partial_\mu \hat{B}_{\beta\alpha} \\
&= \partial_\mu \hat{B}_{\nu\alpha} + \partial_\nu \hat{B}_{\alpha\mu} - \partial_\nu (A_\mu^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_\nu A_\mu^{(1)\beta} \hat{B}_{\alpha\beta} - \partial_\mu (A_\nu^{(1)\beta} \hat{B}_{\beta\alpha}) + \partial_\mu A_\nu^{(1)\beta} \hat{B}_{\beta\alpha} \\
&= \partial_\mu \hat{B}_{\nu\alpha} - \partial_\nu \hat{B}_{\alpha\mu} - \partial_\nu (A_\mu^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_\nu A_\mu^{(1)\beta} \hat{B}_{\alpha\beta} + \partial_\mu (A_\nu^{(1)\beta} \hat{B}_{\alpha\beta}) - \partial_\mu A_\nu^{(1)\beta} \hat{B}_{\alpha\beta} \\
&= \partial_\mu (\hat{B}_{\nu\alpha} + A_\nu^{(1)\beta} \hat{B}_{\alpha\beta}) - \partial_\nu (\hat{B}_{\mu\alpha} + A_\mu^{(1)\beta} \hat{B}_{\alpha\beta}) - \hat{B}_{\alpha\beta} (\partial_\mu A_\nu^{(1)\beta} - \partial_\nu A_\mu^{(1)\beta}) \\
&= F_{\mu\nu\alpha}^{(2)} - \hat{B}_{\alpha\beta} F_{\mu\nu}^{(1)\beta}
\end{aligned} \quad (37)$$

where

$$F_{\mu\nu\alpha}^{(2)} = \partial_\mu A_{\nu\alpha} - \partial_\nu A_{\mu\alpha},$$

and

$$A_{\mu\alpha}^{(2)} = \hat{B}_{\mu\alpha} + B_{\alpha\beta} A_\mu^{(1)\beta}.$$

Finally for $H_{\mu\nu\rho}$ we are just going to use the same technique as before which gave us

$$\begin{aligned}
H_{\mu\nu\rho} &= V_\mu^r V_\nu^s V_\rho^t \hat{V}_r^\mu \hat{V}_s^\nu \hat{V}_t^\rho \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \\
&= \hat{H}_{\mu\nu\rho} - A_\mu^{(1)\alpha} \hat{H}_{\alpha\nu\rho} - A_\nu^{(1)\alpha} \hat{H}_{\alpha\rho\mu} - A_\rho^{(1)\alpha} \hat{H}_{\alpha\mu\nu} + A_\mu^{(1)\alpha} A_\nu^{(1)\beta} \hat{H}_{\alpha\beta\rho} + A_\nu^{(1)\alpha} A_\rho^{(1)\beta} \hat{H}_{\alpha\beta\mu} + A_\rho^{(1)\alpha} A_\mu^{(1)\beta} \hat{H}_{\alpha\beta\nu} \\
&= \partial_\mu B_{\nu\rho} - \frac{1}{2} (A_\mu^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2} (A_\nu^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2} (A_\rho^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)})
\end{aligned} \quad (38)$$

where

$$B_{\mu\nu} = \hat{B}_{\mu\nu} + \frac{1}{2} A_\mu^{(1)\alpha} A_{\nu\alpha}^{(2)} - \frac{1}{2} A_\nu^{(1)\alpha} A_{\mu\alpha}^{(2)} - A_\mu^{(1)\alpha} B_{\alpha\beta} A_\nu^{(2)\beta} \quad (39)$$

5.2 Observation of gauge

Now as in part 4.3 we are going to observe the gauge rule to be sure that every object transform as it should so let's begin with $A^{(2)}$ which should transform with the lie derivative and under the new transformations of $B_{\mu\nu} \rightarrow B_{\mu\nu} + \partial_\mu u \Lambda_\nu$, so let's begin with $A_{\mu\alpha}^{(2)}$

$$\begin{aligned}
\delta A_{\mu\alpha}^{(2)} &= \delta(\hat{B}_{\mu\alpha} + B_{\alpha\beta} A_\mu^{(1)\beta}) \\
&= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta} A_\mu^{(1)\beta}) \\
&= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta}) A_\mu^{(1)\beta} + B_{\alpha\beta} \delta(A_\mu^{(1)\beta}) \\
&= L_\xi \hat{B}_{\mu\alpha} + \partial_\mu \Lambda_\alpha^{(2)} + A_\mu^{(\beta)} L_\xi \hat{B}_{\alpha\beta} + \hat{B}_{\alpha\beta} \partial_\mu \Lambda_\alpha^{(1)} \\
&= L_\xi \hat{B}_{\mu\alpha} + \partial_\mu \Lambda_\alpha^{(2)} + L_\xi(A_\mu^{(\beta)} \hat{B}_{\alpha\beta}) \\
&= \partial_\mu \Lambda_\alpha^{(2)} + L_\xi A_{\mu\alpha}^{(2)}
\end{aligned} \tag{40}$$

so we have the right transformation for this object

Doing the same thing with $B_{\mu\nu}$ we come under the transformation

$$\delta B_{\mu\nu} = \frac{1}{2}(\Lambda^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + \lambda_\alpha^{(2)} F_{\mu\nu}^{(1)\alpha}), \tag{41}$$

wich is exactly the good terme to compensate the additional term for the transformation of $H_{\mu\nu\rho}$

6 Complete reduced action

To recap we have the complete reduced action in the form

$$S = \int dx^D V e^{-\phi} \mathcal{L}, \tag{42}$$

Where we decompose S as $S = S_1 + S_2 + S_3 + S_4$, and we pose

$$\begin{aligned}
S_1 &= \int dx^D V e^{-\phi} (R + g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi), \\
S_2 &= \int dx^D V e^{-\phi} \left(\frac{1}{4} g^{\mu\nu} (\partial_\mu h_{\alpha\beta} \partial_\nu G^{\alpha\beta} - h^{\alpha\beta} h^{\gamma\delta} \partial_\mu B_{\alpha\gamma} \partial_\nu B_{\beta\delta}) \right), \\
S_3 &= \int dx^D V e^{-\phi} \left(-\frac{1}{4} g^{\mu\rho} g^{\nu\lambda} (h_{\alpha\beta} F_{\mu\nu}^{(1)\alpha} F_{\rho\lambda}^{(1)\beta} + h^{\alpha\beta} H_{\mu\nu\alpha} H_{\rho\lambda\beta}) \right), \\
S_4 &= \int dx^D V e^{-\phi} \left(-\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right).
\end{aligned} \tag{43}$$

7 Observation of new symmetry

7.1 O(E,E) group presentation

Now for this part we are going to explore a very important part of physics research nowadays which is symmetry. In our context it will allow us to make equivalence between multiple theory as whole. In the first place we can see that we can have an $SL(E, R)$ symmetry, which is every matrix $E \times E$ matrix with determinant of one. There other symmetry that we can see for example the fact that, B only appear within a derivative mean that we can add a constant to it however we want, and as shown earlier in part 5.2, and in 4.3, we have some gauge freedom that we can use as we please. However, all those symmetry are manifest in high dimensional, what will interest us in the reduction is that it does not create nor destroy any symmetry and if we observe a new symmetry in the reduced form that mean that we have effectively discovered a new symetrie in the unreduced forme. and following the [3] we observe an O(E,E) symmetry

Definition 1 Any matrices M is in $O(E,E)$ if it is a square $E \times E$ matrices and if for the matrices

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We can use this matrices as a metric for the groupe $O(E,E)$ as it has d -1 eigenvalues and d 1 eigenvalues, in a basis rotated from the one with diagonal metric.

the matrices M verify

$$M^T \zeta M = \zeta$$

7.2 Rewriting of the reduced action

We will now define a new matrices M which is

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \quad (44)$$

We will now try to rewrite the action in function of M to make it easier to see. S_1 is a lost cause as neither g or ϕ are contained within M , so let's begin, with S_2 , which can be rewritten in a matricial form :

$$S_2 = \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B),$$

And now we are going to do a few manipulation

$$\begin{aligned} S_2 &= \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} \text{tr} \left(2\partial_\mu h \partial^\mu h^{-1} + 2h^{-1} \partial_\mu B h^{-1} \partial^\mu B \right) \\ &= \frac{1}{8} \text{tr} \left(2\partial_\mu h \partial^\mu h^{-1} - 2\partial_\mu (Bh^{-1}B) \partial^\mu h^{-1} + \partial_\mu (Bh^{-1}) \partial^\mu (Bh^{-1}) + \partial_\mu (h^{-1}B) \partial^\mu (h^{-1}B) \right) \\ &= \frac{1}{8} \text{tr} \left(\partial_\mu (h - Bh^{-1}B) \partial^\mu h^{-1} + \partial_\mu (Bh^{-1}) \partial^\mu (Bh^{-1}) + \partial_\mu (h^{-1}B) \partial^\mu (h^{-1}B) + \partial^\mu h^{-1} \partial_\mu (h - Bh^{-1}B) \right) \\ &= \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) \end{aligned} \quad (45)$$

Now we are going to rewrite S_3 by developing the H partial to get

$$\begin{aligned} S_3 &= -\frac{1}{4} [F_{\mu\nu}^{(1)\alpha} h_{\alpha\beta} F^{(1)\mu\nu\beta} + (F_{\mu\nu\alpha}^{(2)} - B_{\alpha\gamma} F_{\mu\nu}^{(1)\gamma}) h^{\alpha\beta} (F_{\beta}^{(2)\mu\nu} - B_{\beta\delta} F^{(1)\mu\nu\delta})] \\ &= -\frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \end{aligned} \quad (46)$$

Where $\mathcal{F}_{\mu\nu}^i$ is the 2E-component vector of field strengths

$$\mathcal{F}_{\mu\nu}^i = \begin{pmatrix} F_{\mu\nu}^{(1)\alpha} \\ F_{\mu\nu\alpha}^{(2)} \end{pmatrix} = \partial_\mu \mathcal{A}_\nu^i - \partial_\nu \mathcal{A}_\mu^i \quad (47)$$

And finally we go to S_4 firstly let's rewrite $H_{\mu\nu\rho}$

$$\begin{aligned} H_{\mu\nu\rho} &= \partial_\mu B_{\nu\rho} - \frac{1}{2} (A_\mu^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2} (A_\nu^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2} (A_\rho^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)}) \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2} \mathcal{A}_\mu^i \zeta_{ij} \mathcal{F}_{\nu\rho}^j + (\text{cyc.perm}) \end{aligned} \quad (48)$$

We can now rewrite the complete action in terms of M , that gave us:

$$\begin{aligned} S &= \int dx^D V e^{-\phi} [R + g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) - \frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \\ &\quad - \frac{1}{12} (\partial_\mu B_{\nu\rho} - \frac{1}{2} \mathcal{A}_\mu^i \zeta_{ij} \mathcal{F}_{\nu\rho}^j + (\text{cyc.perm})) (\partial^\mu B^{\nu\rho} - \frac{1}{2} \mathcal{A}^{i\mu} \zeta_{ij} \mathcal{F}^{j\nu\rho} + (\text{cyc.perm}))] \end{aligned} \quad (49)$$

7.3 Action invariance under $O(E,E)$

We will now study the effect of the $O(E,E)$ group on this action, firstly as it only acts on the internal index, so the action of the group on the different part of the are

$$g_{\mu\nu} \rightarrow g_{\mu\nu}, \quad (50)$$

$$\phi \rightarrow \phi \quad (51)$$

It is worth noting that for ϕ the whole thing is invariant however each component ($\hat{\phi}$, and $\frac{1}{2} \log(h)$) on there one are not invariant to $O(E,E)$ group, which is why we needed to create the shifted dilaton.

We also need to define the action of the group on the two form, and for $A^{(2)}$, when they have internal index, so we have:

$$A^i \rightarrow \Omega_j^i A^j \quad (52)$$

$$M_{ij} \rightarrow \Omega_i^l M_{lk} \Omega_j^k \quad (53)$$

Here $\Omega \in O(E, E)$. We now show that $M \in O(E, E)$:

$$\begin{aligned} M^T \zeta M &= \begin{pmatrix} h^{-1} & Bh^{-1} \\ -h^{-1}B & h - Bh^{-1}B \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} Bh^{-1} & h^{-1} \\ h - Bh^{-1}B & -h^{-1}B \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \zeta, \end{aligned}$$

we can now verify that every part of the action is invariant under the action $O(E, E)$. For S_1 the invariance is manifest as nothing depend on the internal space. For S_2 we can use the ciclearisation of the trace and the action of $O(E, E)$ on M which is with $\Omega \in O(E, E)$

$$M \rightarrow \Omega M \Omega^T$$

to make the invariance of $O(E, E)$ more visible. In S_3 the technic Now as $\Omega^T \zeta \Omega = \zeta$ the seconds termes is $O(E, E)$ invariant. And the for the first terme is invariant if we require that $B_{\nu\rho}$ does not transfrom. This demonstrates that the action hides a non-trivial symmetry, providing a useful framework for comparing equivalent theories. For S_3 we have using (52)

$$S_{3'} = -\frac{1}{4} \mathcal{F}_{\mu\nu}^i \Omega_i^l \Omega_n^i (M^{-1})_{ij} \Omega_m^j \Omega_j^k \mathcal{F}^{\mu\nu j} \quad (54)$$

$$= -\frac{1}{4} \mathcal{F}_{\mu\nu}^i \Omega_j^k \eta_{kn} \Omega_n^i (M^{-1})_{ij} \Omega_m^j \eta_{ml} \Omega_i^l \mathcal{F}^{\mu\nu j} \quad (55)$$

$$= -\frac{1}{4} \mathcal{F}_{\mu\nu}^i \eta_{ji} (M^{-1})_{ij} \eta_{ij} \mathcal{F}^{\mu\nu j} \quad (56)$$

$$= -\frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \quad (57)$$

$$= S_3 \quad (58)$$

WE can redo this for S_4 and get the same result. We have then proven that the whole action is $O(E, E)$ invariant. This demonstrates that the action hides a non-trivial symmetry, providing a useful framework for comparing equivalent theories.

8 Conclusion

During this intership we have shown two inportants things, firtly that if we take an universe with $4 + E$ dimension, the reduced action in 4 dimensions is going to be the classical Einstein action, coupled with the maxwell action. After that we have shown that if we use the NS-NS action, there was already in high dimension an $SL(E, R)$ invariance, but that under the reduced Action appear the $O(E, E)$ symmetry wich allow us to make equivalence between multiple theory.

Now if we had to extend this project to be more complete is to do general kuluza-klein reduction with a depence to the internal space [1]. We could also search for more general hidden symmetry like the $O(E, E + n)$ [3].

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A spin connexion, vielbein and flatspace

In general relativity we often encounter difficult to handle metrics such as the Kerr-Newman metric. A good idea we can have is to then work in an orthonormal basis $\{e_\mu^a\}$, which follow:

$$g_{\mu\nu} V_a^\mu V_b^\nu = \eta_{ab} \quad (59)$$

where g is the original metric and η is the Minkowski metric.

We call this basis the vielbein. In this basis we will use the Minkowski metric to raise and lower indices. Of course we can define the dual of the vielbein otherwise called the inverse vielbein with

$$V_\nu^a V_b^\nu = \delta_b^a \quad (60)$$

and of course that mean that we can go back with to the original metric with

$$g_{\mu\nu} = \eta_{ab} V_\mu^a V_\nu^b \quad (61)$$

Of course we can also change the basis any random vector, form or tensor with this, the method is to simply do

$$T^a = V_\mu^a T^\mu \quad (62)$$

$$T_a = V_a^\mu T_\mu \quad (63)$$

$$T_b^a = V_\mu^a V_b^\nu T_\nu^\mu \quad (64)$$

Now in this internship we will not be in four dimensions, but rather in 4 time-space dimension, with E supplementarie space dimensions. And to pass from 4 + E total dimension to the other thing we use the vielbein in 4+E dimension the the 4 dimensions vielbein like this:

$$T^\mu = V_r^\mu \hat{V}_{\hat{\mu}}^r \hat{T}^{\hat{\mu}} \quad (65a)$$

$$T^\alpha = V_a^\alpha \hat{V}_{\hat{\mu}}^a \hat{T}^{\hat{\mu}} \quad (65b)$$

$$T_\mu = V_\mu^r \hat{V}_r^{\hat{\mu}} \hat{T}_{\hat{\mu}} \quad (65c)$$

$$T_\alpha = V_\alpha^a \hat{V}_a^{\hat{\mu}} \hat{T}_{\hat{\mu}} \quad (65d)$$

$$(65e)$$

Now of course other cange need to happen to the classical transformation for this to work, such as the classical christofells connexion which does not work anymore in this context so we introduce the spin(or latin) connexion whith

$$\nabla_\mu X_b^a = \partial_\mu X_b^a + \omega_\mu^a{}_c X_b^c - \omega_\mu^c{}_b X_c^a \quad (66)$$

Of course we can still make link with the levi-civita conexion with the equations

$$\Gamma_{\mu\lambda}^\nu = V_a^\nu \partial_\mu V_\lambda^a + V_a^\nu V_\lambda^b \omega_\mu^a{}_b, \quad (67)$$

$$\omega_\mu^a{}_b = V_\nu^a V_b^\lambda \Gamma_{\mu\lambda}^\nu - V_b^\lambda \partial_\mu V_\lambda^a, \quad (68)$$

With those relation we can do prove the anti-symetrie of the last two index of the spin connexion

$$\begin{aligned} \nabla_\mu \eta_{ab} &= \partial_\mu X_{ab} - \omega_\mu^a{}_c \eta_{cb} - \omega_\mu^c{}_b \eta_{ac} \\ &= -\omega_{\mu ab} - \omega_{\mu ba} \\ &= 0 \end{aligned} \quad (69)$$

so in effect we have

$$-\omega_{\mu ab} = \omega_{\mu ba} \quad (70)$$

B lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned} \delta S &= \delta \int \mathcal{L} dx^\mu \\ &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \partial_\nu \eta_\rho dx^\mu \\ &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho + \underbrace{\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho \right)}_{\text{border terme equal to zero}} dx^\mu \\ &= 0 \end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \quad (71)$$

which is in fact the equation of mouvement for the field

C General relativty 101

Definition 2 Let V be an R -vector space of dimension N , we define the tensor of order $(r, s) \in \llbracket 0, 1 \rrbracket^2$ on $V^{(r,s)} \stackrel{def}{=} V^{\otimes r} \otimes V^{\otimes s}$ as a being a multilinear application on $V^{\star r} \times V^s$ to $\mathbb{R}$

Definition 3 We will call n -dimension ($n \in \mathbb{N}^*$) manifold every Hausdorff topological space $(\mathcal{M}, \tau)$ locally homeomorphic to $\mathbb{R}^n$)

Definition 4 Let $(\mathcal{M}, g)$ be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form $g_{\mu\nu}$ on the tangent space $T_p\mathcal{M}$ at each point $p \in \mathcal{M}$, such that

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 5 In general in RG we use the christoffels connexion as it is the sole one wich is mtreci and torsion free. We define it as being

$$\text{nabla}_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\sigma}^\nu V^\sigma \quad (72)$$

where the christoffels symbols are

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \quad (73)$$